

ALLOYS & COMPOUNDS

CHAPTER 7

INTRODUCTION

- An **alloy** is a mixture of two elements, one of which is a metal.
- **Alloys** often have properties that are different to the metals they contain. This makes them more useful than the pure metals alone. For example, **alloys** are often harder than the metal they contain
- They have properties of metals but are not elements

Examples of alloys

- **Bronze-** is a mixture of copper and tin
- Used to make statues, medals, coins, decoration pieces and household utensil
- **Brass** – an alloy of copper and zinc
- Used to make parts of machinery, musical instruments and ornamental objects
- **Stainless steel-** an iron alloy of chromium and nickel
- Used in making cutlery, household utensils, automobile parts and surgical instruments
- **Gold-copper** alloy- used to make durable jewellery

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- **Duralumin** – is an alloy of aluminium, copper, magnesium and manganese
- This alloy is more strong and lighter than aluminium itself
- It is used in manufacture of airplanes and aircrafts
- New Samsung notebook Pc is made of duralumin

Compounds

- Compounds combine in a fixed proportion
- Their properties are different from their constituent elements as well as their MP and BP
- Cannot be separated by any physical means e.g water
- **Water** – contains two hydrogen and one oxygen atoms i.e 2:1
- **Hydrogen peroxide** (H_2O_2)- a toxic liquid with two molecules of hydrogen and two molecules of oxygen. i.e ratio 2:2
- **Sodium chloride (NaCl)**- a common table salt.
- Contains sodium (metal) and chlorine (non-metal)
- Used in manufacturing of glass, soap

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- **Hydrochloric acid (HCl)**- is a mineral acid and highly corrosive
- Used for cleaning washbasins, floor tiles and metallic surfaces
- Stomach contains small amount of HCl that helps in digestion and killing of microorganisms that might be present in the food
- **Limestone and marble**- are forms of calcium carbonates
- White marble is used to make statues and monuments
- Ground calcium carbonate is used as a coating pigment for paper because of its high brightness, and also in paints
- Limestone is used to make slaked lime used for whitewashing walls

Types of compound

1. **Organic compounds**- are found in living organism
 - Have hydrogen, carbon and oxygen elements and small amounts of nitrogen and sulphur
 - Examples include
 - **Hydrocarbons** – these contains hydrogen and carbon only e.g Coal, Petroleum, natural gas
 - **Carbohydrates, proteins, fats**- contains hydrogen, carbon and oxygen atoms in proportions.
2. **man-made materials**- are inorganic compounds that are manufactured. Example, plastics, cement, medicines, soaps, glass and detergents

Differences between alloys and compounds

Alloys	Compound
Homogenous mixture of metal with another or nonmetal	Combination of two elements in a fixed ratio
Does not involve any chemical reaction or bond formation	Involves chemical reactions forms new bonds
One element must be a metal	Two or more non-metals can combine

Note (medicine)

- ***Antacids***- used to relief stomach acidity caused by excessive acid production. E.g milk of magnesia
- ***Analgesics*** (painkillers) - give relief from pain e.g aspirin
- ***Antipyretic***- helps to bring down fever and reduce pain e.g paracetamol
- ***Antiseptics***- used to prevent growth of microorganism in wounds
- ***Antibiotics***- used to inhibit the growth of other diseases causing microorganism
- ***Anti-histamines***- used to treat allergies, hypersensitive reactions and colds

CONCLUSION

THANK YOU MY DEAR STUDENTS

LOVE YOU

Glory be to Allah the GOD of ALL